

AI-Based SOC Triage Tool Integrated with Splunk

Abstract

This project proposes an AI-based SOC triage tool integrated with Splunk to automatically classify security alerts into True Positive and False Positive. It enhances detection accuracy, reduces analyst workload, and improves response efficiency through automation and intelligent decision-making.

Introduction

Security Operations Centers (SOCs) play a critical role in monitoring and defending organizational assets. However, the increasing number of alerts generated by security tools creates major operational challenges for analysts.

Problem Statement

SOC analysts face alert overload, where thousands of alerts are generated daily. A high percentage of these alerts are false positives, leading to wasted effort, delayed responses, and potential oversight of real threats.

Research Gap

1. Lack of automated alert classification in SIEM tools such as Splunk.
2. Heavy reliance on manual triage processes.
3. No intelligent decision support for junior analysts.
4. Limited integration of AI in real-time SOC workflows.
5. Inefficient prioritization of alerts based on risk context.

Proposed Solution

The solution is an AI-powered triage tool that integrates with Splunk, automatically analyzes alerts, classifies them as True/False positives, assigns risk scores, and recommends response actions.

System Architecture

- Data Source: Windows logs and Sysmon logs.
- SIEM: Splunk collects and indexes logs.
- Integration Layer: Splunk API retrieves alerts.

- Processing Engine: Python-based AI module.
- Output: Streamlit dashboard for visualization.

Detailed Workflow

1. Attacks simulated from Kali Linux.
2. Logs generated on Windows system.
3. Splunk ingests logs and creates alerts.
4. AI tool retrieves alerts through API.
5. Alert parsing and feature extraction.
6. AI model classifies alerts.
7. Results displayed on dashboard.

AI Model Design

The model uses features such as Event ID, frequency of events, login patterns, IP reputation, and timestamps. A hybrid approach is used:

- Rule-based for initial classification.
- Machine learning (e.g., Random Forest) for advanced classification.

Splunk Integration

The tool connects to Splunk using REST API to retrieve alerts in real time. Example:

- Endpoint: /services/search/jobs/export
- Query: search index=wineventlog EventCode=4625

Dashboard Features

- Alert table with classification results.
- Risk distribution charts.
- Timeline of alerts.
- Search and filtering capabilities.

Evaluation Metrics

The system is evaluated using:

- Accuracy
- Precision
- Recall
- False Positive Rate

Expected Results

- Reduction in false positives.
- Faster triage process.
- Improved SOC efficiency.
- Better decision-making support.

Conclusion

This project bridges a key gap in SOC operations by integrating AI with SIEM systems to automate alert triage. It provides a scalable and intelligent solution for modern cybersecurity challenges.

Future Work

- Integration with SOAR tools for automated response.
- Use of deep learning models.
- Real-time threat intelligence enrichment.
- Deployment in enterprise environments.